

SPRINGFIELD-BRANSON, MO, USA

WMO: 724400

Lat: 37.2347N

Lon: 93.4014W

Elev: 1280

StdP: 14.03

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13995

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.0	12.6	-2.4	5.1	10.0	2.7	6.6	15.4	26.6	42.7	24.6	41.1	8.1	340	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	95.3	74.1	92.3	74.4	89.8	74.1	77.8	88.9	76.8	87.7	75.5	86.0	8.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.6	136.0	83.5	73.5	130.8	82.5	72.4	126.0	81.3	42.1	88.9	41.0	87.5	39.9	86.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	20.6	18.7	-0.6	98.5	5.8	3.3	-4.7	100.9	-8.1	102.8	-11.4	104.7	-15.5	107.1
			-1.4	79.4	5.5	1.4	-5.4	80.3	-8.6	81.1	-11.7	81.9	-15.7	82.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.0	33.7	38.1	47.0	56.7	65.6	74.4	78.5	77.7	69.9	58.3	46.7	36.8
	DBStd	18.06	11.89	11.47	10.89	9.02	7.84	5.65	4.96	5.38	7.57	9.05	9.76	10.48
	HDD50	1733	518	359	186	39	3	0	0	0	0	28	174	426
	HDD65	4369	970	753	562	273	93	6	0	1	38	246	551	876
	CDD50	4306	13	26	93	240	485	733	884	859	598	285	74	16
	CDD65	1466	0	0	4	24	110	290	419	395	185	37	2	0
	CDH74	13683	0	4	37	191	814	2629	4274	3856	1588	276	14	0
	CDH80	5532	0	0	2	24	185	997	1934	1761	574	55	0	0
Wind	WSAvg	9.2	10.1	10.3	10.9	10.8	9.2	8.4	7.1	7.2	7.8	9.0	9.9	9.9
Precipitation	PrecAvg	43.4	2.0	2.3	3.6	4.4	4.7	4.8	3.4	3.6	4.4	3.5	3.6	3.0
	PrecMax	62.0	6.8	7.6	9.4	12.0	13.4	13.4	9.2	8.6	17.5	10.0	12.2	11.5
	PrecMin	27.4	0.1	0.1	0.9	0.2	1.5	0.2	0.3	0.4	0.1	0.4	0.2	0.4
	PrecStd	8.4	1.4	1.6	2.0	2.4	2.2	2.7	2.1	1.9	3.3	2.2	2.6	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.4	72.6	78.1	83.1	87.5	94.1	98.9	99.1	94.1	85.7	75.7	67.8
		MCWB	56.3	57.7	62.0	65.3	71.9	74.6	72.9	73.3	72.2	69.9	62.6	59.1
	2%	DB	62.4	66.0	73.6	79.3	84.5	90.9	95.3	95.7	89.8	80.8	70.1	62.4
		MCWB	53.2	54.5	59.4	63.3	70.2	74.2	74.5	74.2	72.1	66.4	59.3	55.3
	5%	DB	57.2	61.5	68.8	75.6	81.9	88.4	92.2	92.5	86.5	76.8	66.1	57.3
		MCWB	49.7	52.3	56.6	61.9	69.3	73.6	75.0	74.1	71.3	64.1	57.7	50.8
	10%	DB	51.4	56.7	64.6	72.1	78.9	86.0	89.5	89.2	83.0	72.8	62.3	52.8
		MCWB	45.4	48.9	54.3	60.1	67.8	72.8	75.1	73.7	69.1	62.9	54.8	47.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.9	61.7	64.7	68.4	74.4	78.1	79.4	78.7	75.9	72.2	65.5	61.5
		MCDB	65.3	69.1	74.8	78.3	84.3	89.1	90.0	90.6	86.8	81.4	72.4	66.5
	2%	WB	55.8	57.9	61.6	66.3	72.7	76.6	78.1	77.4	74.3	69.5	62.1	56.8
		MCDB	60.3	63.8	71.3	75.2	81.7	87.4	88.9	89.1	85.3	76.2	67.6	61.1
	5%	WB	50.7	53.6	58.8	64.3	70.8	75.1	77.1	76.2	72.9	67.4	59.2	52.2
		MCDB	55.9	59.1	66.6	72.2	79.4	85.3	87.8	87.5	83.2	73.5	64.5	55.8
	10%	WB	45.3	48.9	55.8	62.1	69.0	73.9	76.1	74.9	71.4	64.6	56.0	47.8
		MCDB	51.8	56.5	63.0	69.9	76.8	83.4	86.3	85.6	80.1	70.8	61.4	51.8
Mean Daily Temperature Range	5% DB	MDBR	18.9	19.7	21.0	21.1	19.1	18.9	19.1	19.9	20.6	20.7	19.7	17.7
		MCDBR	26.1	26.5	26.1	24.6	20.8	21.1	22.3	23.7	22.5	23.3	24.1	23.8
		MCWBR	17.8	17.7	15.4	13.6	10.4	8.6	7.3	7.5	8.8	11.8	15.9	17.0
	5% WB	MCDBR	22.5	23.5	22.0	20.6	19.0	19.1	19.1	19.9	19.5	19.1	21.0	20.9
		MCWBR	17.3	17.8	14.8	13.4	10.5	9.0	7.9	7.8	8.5	11.1	16.0	16.9
Clear-Sky Solar Irradiance	taub	0.299	0.307	0.339	0.376	0.414	0.422	0.440	0.438	0.380	0.340	0.307	0.291	
	taud	2.467	2.451	2.406	2.321	2.272	2.278	2.275	2.257	2.392	2.483	2.520	2.513	
	Ebn at Noon	284	295	294	287	276	273	267	265	276	279	279	279	
	Edh at Noon	26	30	35	40	43	43	42	42	35	29	25	23	
All-Sky Solar Radiation	RadAvg	750	938	1255	1597	1789	2050	2003	1810	1551	1136	801	632	
	RadStd	73	125	77	93	142	163	127	118	119	131	86	75	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page